Initial Ratio of Strontium 87 to Strontium 86, Whole-Rock Age, and Discordant Biotite in the Monteregian Igneous Province, Quebec[1]

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Abstract. Rb-Sr isotopic analyses of sixteen whole-rock samples from ten localities in the Monteregian igneous province, Quebec, show that the initial Sr^{87}/Sr^{86} is 0.704, which is in essential agreement with our current estimate for oceanic and nonorogenic continental basalt. An isochron plot of nordmarkite and essexite from adjoining localities, Brome and Shefford, gives $(Sr^{87}/Sr^{86})_0 = 0.7039$ and an age of 124 m.y. (early Cretaceous). The former agrees with the over-all estimate; the latter is in approximate agreement with four of five K-Ar biotite ages but is significantly higher than five Rb-Sr biotite analyses. A radiogenic Sr^{87} development plot of biotites and whole rocks from Brome shows that the biotites from the comagnetic nordmarkite and essexite have comparable ages (84.4 and 88.3 m.y., respectively), thus indicating a postcrystallization Sr diffusion episode late in the Cretaceous.

Introduction. The general geology and petrography of the isolated igneous hills which extend in an irregular line east from Montreal across the St. Lawrence lowland are too well known to need extended description (Figure 1). *Dresser and Denis* [1944] summarized the early literature and have described the contrasting plutonic types, in general varieties of alkali-gabbro and syenite, which make up the hills. A comagmatic origin and, from field relations, an age not greater than Devonian has been generally assumed. The present investigation adds nothing of interest to the mineralogy and general geology of the province; its purpose is primarily to find the magnitude of the critical ratio Sr^{87}/Sr^{86} at zero time in the crystallization of the Monteregian parent magma and to measure later changes in this ratio from which the age of the rocks and minerals can be computed. Preliminary data have been reported by *Faure and Hurley* [1963] and *Fairbairn et al.* [1961*a*].

The analytical procedures used are standard and need no elaboration here. Sr^{87}/Sr^{86} analyses are repeatedly checked against the value obtained with a standard Eimer and Amend Sr carbonate (lot 492327). Most laboratories agree that this standard value is 0.708 if corrected in accordance with an assumed value of 0.1194 for the Sr^{86}/Sr^{88} ratio. Experience with replicate analyses indicates that the precision error of a single Sr^{87}/Sr^{86} determination is about ± 0.0005. Rb and Sr analyses were made by the isotope dilution method, with the exception of those samples (see Table 1) having an extremely small Rb/Sr ratio. In such cases, the approximate X-ray ratio is satisfactory.

Initial Sr^{87}/Sr^{86} of the Monteregian parent magma. An estimate of the initial (precrystallization) Sr^{87}/Sr^{86} can be made either from samples containing little or no Rb or by extrapolation to zero Rb of the isochron drawn on a Sr^{87}/Sr^{86}-Rb^{87}/Sr^{86} plot [*Nicolaysen,* 1961]. Both approaches are used here with generally concordant results.

Table 2 is composed of selected data from the comprehensive Table 1; only those samples having Rb/Sr $\lesssim$ 0.11 are used. If an approximate age of 100 m.y. is assumed, the corrected values of Sr^{87}/Sr^{86} should then give a mean value of the critical initial ratio for the source magma of the Monteregian intrusives. This has been computed as 0.7040 ± 0.0002. (An average of 0.705 was published in our 1962 Annual Report to the Atomic Energy Commission. Analyses of all samples performed again under more uniform and critical conditions now average 0.704.) The maximum correction for any of the individually

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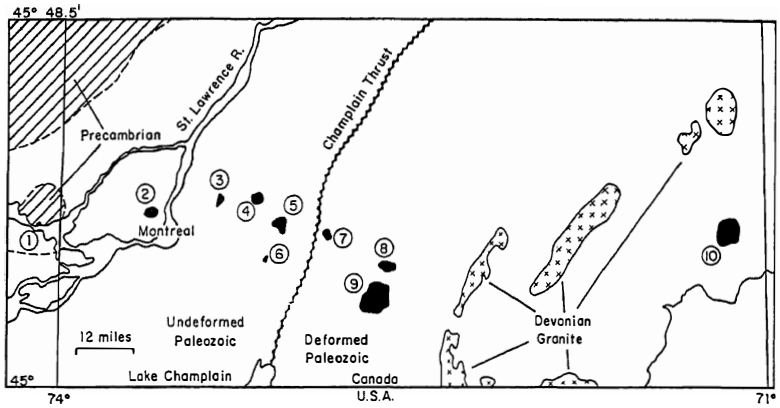


Fig. 1. Sketch map of the Monteregian Hills, including Oka and Megantic as new members. Numbers refer to locations given in Table 1.

measured Sr^{87}/Sr^{86} ratios is 0.0006 (St. Bruno). The ratio 0.704 is identical with that obtained by *Faure and Hurley* [1963] for alkaline oceanic basalt, and it compares favorably with 0.705 obtained for nonorogenic continental types. (As a result of improved procedures and re-analysis of key samples, the values published by Faure and Hurley, averaging 0.707 to 0.708, should all be lowered by about 0.003.) As the spread of the corrected ratios in Table 2 (0.7049 to 0.7032) is greater than the ±0.0005 error of a single analysis, the variation is believed due in part to small initial Sr^{87}/Sr^{86} differences in the magma and not to experimental error only.

The isochron approach for the initial Sr^{87}/Sr^{86} ratio is shown in Figure 2. Four whole-rock samples, two each from Brome and Shefford, provide closely collinear points over a satisfactory range of Rb^{87}/Sr^{86} and Sr^{87}/Sr^{86} values. A line has been fitted to these points by the method of least squares. Extrapolation of this isochron line to zero Rb^{87}/Sr^{86} gives 0.7039 for the initial Sr^{87}/Sr^{86} ratio. The uncertainty in this value by least-squares methods is ±0.0001(5). Summarizing the data of Table 2 and Figure 2, we conclude that $(Sr^{87}/Sr^{86})_0$ is 0.704 within limits considerably less than ±0.001.

Whole-rock age. A complete age study has not been made. Whole-rock samples of adequate Rb/Sr range were collected only from Brome and Shefford. The slope of the isochron in Figure 2 gives 124 m.y. as the age for these two localities. The error has been computed by least-squares methods to be ±3 m.y. These are closely associated localities (numbers 8 and 9 in Figure 1) and may indeed form a single body [*Dresser and Denis*, 1944]. On the basis of *Holmes'* [1960] time scale an early Cretaceous age is indicated.

A single analysis of granite from Megantic gives 113 m.y., with an error (±5 m.y.) too small for probable age correlation with the Brome-Shefford isochron (compare data in Figure 2). A special study of Megantic should be made to resolve this apparent age difference. Extensive sampling of the syenitic phases of other Monteregian intrusives would also be rewarding if an adequate range of the Rb/Sr values could be found.

Biotite ages. As several K-Ar biotite ages have already been computed from Monteregian rocks (M.I.T. unpublished data; *Lowdon* [1961]), a few Rb-Sr biotite ages were determined for comparison. The data are summarized in Tables 3, 4, and 5. Inspection of the Rb-Sr ages (Table 4) shows that there is little internal agreement and that all of the numbers are smaller than the Brome-Shefford whole-rock age of 124 m.y. Additional comparison with the K-Ar data (Table 5) shows agreement only for the Mount Johnson example. With the exception of Oka, however, all the K-Ar ages are in approximate agreement with the Brome-Shefford isochron of 124 m.y. Considering all data, we feel that the emplacement of the Monteregian bodies in Cretaceous time is established.

Discordant biotite ages from Brome. Complete Sr isotopic data are available for biotite concentrated from the Brome nordmarkite and essexite. A Sr^{87} development diagram (Figure 3)

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TABLE 1. Analytical Data

(Numbers 1 to 10 refer to locality numbers on map.)

Locality, Rock Type, and Sample Number	Rb, ppm by weight	Sr, ppm by weight	Rb/Sr, by weight	Rb^{87}/Sr^{86}, atomic	Sr^{87}/Sr^{86}, atomic	Age, m.y.	Sr^{87}/Sr^{86}, initial
	739	357	2.07				
	757	355	2.13				
1. Oka	718	340	2.11				
B3253 (biotite in carbonatite)	738	351	2.10	6.08	0.7142	114 ± 7	0.7040[a]
C3253 (calcite in carbonatite)			0[b]		0.7032		0.7032
C4584 (calcite in carbonatite)			0[b]		0.7032		0.7032
2. Mount Royal							
R3069 (tinguaite)			0.037[b]	0.107	0.7045		0.7043
3. St. Bruno							
R4704 (gabbro)			0.113[b]	0.327	0.7046		0.7040
4. St. Hilaire							
R4270 (essexite)			0.054[b]	0.156	0.7038		0.7036
B4281b (biotite concentrate from essexite)	650	84.3 (0.269 Sr^{87}*)				99.5 ± 4	0.7040[a]
5. Rougement							
R4703 (peridotite)			0[b]		0.7047		0.7047
6. Johnson							
B4180 b (biotite concentrate from essexite)	582	91.8	6.34	18.4	0.7343	111 ± 6	
R4180 (essexite)			0.04[b]	0.12	0.7033		0.7031
R4268 A (pulaskite)			0.041	0.12	0.7042		0.7040
7. Yamaska							
R4179 (yamaskite)	59.6	1646	0.036	0.105	0.7049		0.7047
R4273 B (syenite)	77.9	1031	0.0756	0.219	0.7056		0.7052
R4273 A (gabbro)			0.01[b]	0.03	0.7048		0.7047
8. Shefford							
R4269 A (nordmarkite)	144	399	0.360	1.042	0.7057		
R4181 (nordmarkite)	140	63.7	2.20	6.37	0.7156		
9. Brome							
B4564 (biotite in nordmarkite)	248	6.81 6.86	36.4 36.1	106.6 106.1	0.8385	84.4 ± 2	
R4564 (nordmarkite)			1.27[b]	3.68	0.7107		0.7040[a]
B4768A (biotite in essexite)	283	28.1	10.1	29.3	0.7424	88.3 ± 3	
R4768A (essexite)			0.047[b]	0.136	0.7047		0.7046
R4768B (essexite)			0.020[b]	0.060	0.7034		0.7034
10. Megantic							
B4196 (biotite in granite)	1402 1412	8.92 8.69	160		(0.570 Sr^{87}*) (0.598 Sr^{87}*)	98 102	[c]
R4196 (granite)	259	57.9	4.47	12.96	0.7256	113 ± 5	

[a] Average of Table 2.

[b] X-ray spectrographic determination, approximate only, based on one scan.

[c] Initial Sr^{87} abundance not critical because of unusually large Rb/Sr.

Note: $\lambda_{Rb} = 1.47 \times 10^{-11}\ yr^{-1}$.

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TABLE 2. Summary of Measured and Corrected (Initial) Sr^{87}/Sr^{86} Data

Location, Rock Type, and Map Number	Approximate Rb/Sr (X-ray), lowest ratios only	Sr^{87}/Sr^{86}, measured	$(Sr^{87}/Sr^{86})_0$, based on 100 m.y.
1. Oka			
Carbonatites	0	0.7032	0.7032 (mean)
	0	0.7032	
2. Mount Royal			
Tinguaite dike	0.04	0.7045	0.7043
3. St. Bruno			
Gabbro	0.11	0.7046	0.7040
4. St. Hilaire			
Essexite	0.05	0.7038	0.7036
5. Rougement			
Peridotite	0	0.7047	0.7047
6. Johnson			
Essexite	0.04	0.7033	0.7036 (mean)
Pulaskite	0.04	0.7042	
7. Yamaska			
Yamaskite	0.04	0.7049	
Syenite	0.08	0.7056	0.7049 (mean)
Gabbro	0.01	0.7048	
9. Brome			
Essexites	0.05	0.7047	0.7040 (mean)
	0.02	0.7034	
		Mean $(Sr^{87}/Sr^{86})_0$ = 0.7040 ± 0.0002	

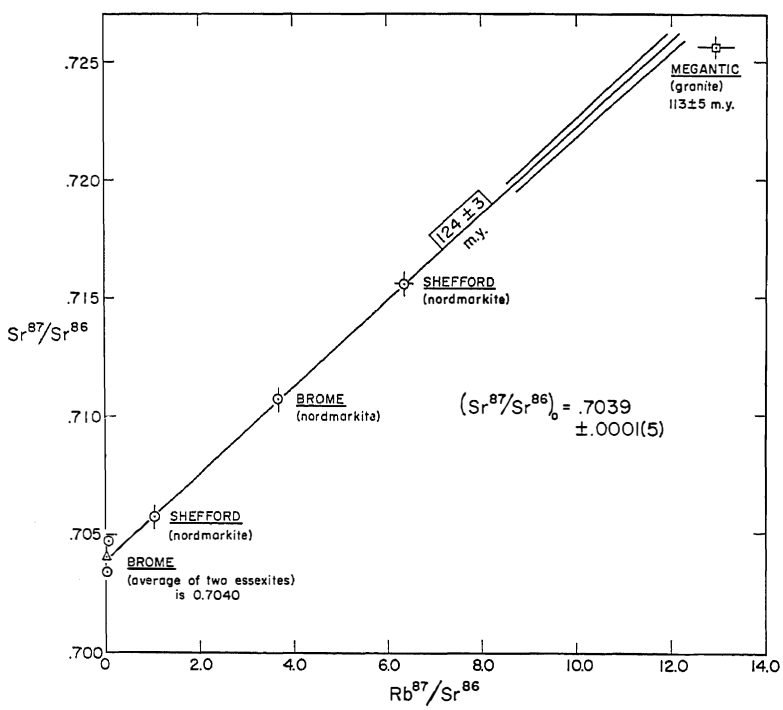


Fig. 2. Whole-rock isochron for nordmarkite and essexite from Brome and Shefford.

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TABLE 3. K–Ar Biotite Analyses

Sample Number, Rock, and Locality	K, %	Ar^{40}/K^{40}	Air Correction, %	Age, m.y.
Johnson				
B3068 (essexite)	5.52	0.00665	44	110 ± 10
Oka				
B3253 (carbonatite)	6.99	0.00576	7.8	95 ± 5
Brome				
GSC59–92 (nordmarkite, *Lowden et al.* [1961])	6.92	0.00735	37	122 ± 10
Megantic				
GSC60–117 (gabbro, *Lowden et al.* [1961])	7.42	0.00760	0	126 ± 10
Megantic				
GSC60–118 (granodiorite, *Lowden et al.* [1961])	7.11	0.00697	12	115 ± 9

Constants: $\lambda_e = 0.585 \times 10^{-10}$ yr^{-1}.
$\lambda_{total} = 5.30 \times 10^{-10}$ yr^{-1}.

has therefore been prepared to illustrate the nature of the discordant ages. The nordmarkite and essexite development lines intersect at 0.704 and 124 m.y., as already determined from the Brome-Shefford isochron (Figure 2). The intersection of each biotite development line with the line for its own host rock then gives the time when each biotite began to accumulate radiogenic Sr. On the assumption of a closed system, for which the whole-rock isochron provides solid support, the much smaller biotite ages are assumed to result from a postcrystallization diffusion episode which reduced their initially high Sr^{87} concentration to the level of their environment [*Compston and Jeffery*, 1959; *Fairbairn et al.*, 1961*b*]. Since the completion of this event, approximately 86 m.y. ago, accumulation of radiogenic Sr is assumed to have been uninterrupted. The diagram contrasts the whole-rock development lines, undisturbed since crystallization of the parent magma 124 m.y. ago, with the steeper biotite

TABLE 4. Rb–Sr Biotite Analyses

Locality and Rock Type	Rb, ppm by wt.	Sr, ppm by wt.	Rb^{87}/Sr^{86}	Sr^{87}/Sr^{86}	Age, m.y.
Oka					
(carbonatite)	738	351[a]	6.08	0.7142	114 ± 7 (mean of 3)
Johnson[b]					
(essexite)	582	91.8	18.4	0.7343	111 ± 6
Megantic					
(granite)	1407	8.80 (0.584 Sr^{87*})			100 ± 1 (mean of 2)
St. Hilaire[b]					
(essexite)	650	84.3 (0.269 Sr^{87*})			99.5 ± 4
Brome					
(nordmarkite,	248	6.83	106	0.8385	84.4 ± 2
essexite)	283	28.1	29.3	0.7424	88.3 ± 3

[a] The high Sr content may reflect in part a 0.3 per cent Sr concentration in this carbonatite.
[b] These samples are diluted in varying degree with rock fragments. Owing to low concentration of biotite in the rocks and the character of the foreign particles, further purification was found to be impracticable.

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TABLE 5. Comparison of Rb–Sr and K–Ar Biotite Ages

Locality and Rock Type	Rb–Sr, m.y.	K–Ar, m.y.	Comment
Oka (carbonatite)	114 ± 7	95 ± 5	Discordant
Johnson (essexite)	111 ± 6	110 ± 10	Accordant; reasonable agreement with the Brome-Shefford isochron (124 m.y.)
Megantic (granite)	100 ± 1	115 ± 9 (G.S.C.)	Discordant; K–Ar agrees with 113 m.y. age of single granite analysis
Megantic (gabbro)		126 ± 10 (G.S.C.)	
St. Hilaire (essexite)	99.5 ± 4		
Brome (nordmarkite, essexite)	84.4 ± 2 88.3 ± 3	122 ± 10 (G.S.C.)	Whole-rock Rb–Sr Age is 124 m.y. (Figure 2)

development lines, which originated at a much later time.

An alternative hypothesis for low biotite ages would involve the addition of Rb to the biotite rather than the removal of Sr. *Kulp and Engels* [1963] have advocated this hypothesis for biotites in open-system metamorphic rocks as part of a base-exchange reaction to which biotite is particularly susceptible. As these conditions are not met in the Monteregian rocks, the hypothesis of Sr^{87} loss from biotite by thermal diffusion, with over-all retention by the rock, seems more plausible.

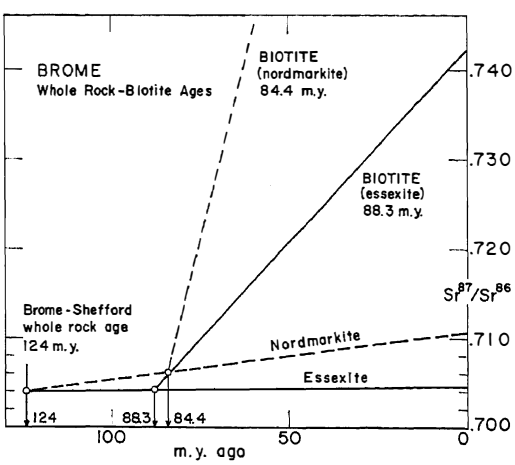


Fig. 3. Sr^{87} development lines for two whole rocks (Brome and Shefford) and for biotite contained in them.

Concluding remarks. All evidence relating to initial Sr^{87}/Sr^{86} points to 0.704 as the critical mean, which is in essential agreement, as already noted, with both oceanic and nonorogenic continental basalt. The carbonatite at Oka (0.7032) is also identical with the mean established by *Powell et al.* [1963] for carbonatites in general (0.7035). As carbonatites are typically associated with alkaline igneous rock types, the Sr^{87}/Sr^{86} value of 0.704 for the Monteregian intrusives, alkaline oceanic basalt, and carbonatites may lend weight to the suggestion of *Yoder and Tilley* [1962] that alkaline basalt magmas may originate in the mantle at slightly greater depth than tholeiitic basaltic magma.

The Brome-Shefford whole-rock age of 124 m.y.—early Cretaceous according to *Holmes'* [1960] time scale—may not represent the time of intrusion of all the other bodies. Preliminary evidence suggests that Megantic may have been emplaced somewhat later in the Cretaceous (113 m.y.).

The discordant age of the Brome biotites has been correlated with a Sr diffusion event about 86 m.y. ago. If adequate isochron plots can be established for other intrusive centers in the Monteregian group, the resultant over-all comparison of whole-rock and biotite ages would make a welcome addition to the present study.

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Appendix 1. Earlier Age Estimates

Osborne [*Dresser and Denis,* 1944] noted that pleochroic halos surrounding zircon and sphene were poorly developed in the Monteregian intrusives and suggested a Tertiary age for the province. This has proved to be not far from correct. Previously, in 1936, Urey [*Dresser and Denis,* 1944] reported a helium age of 57 m.y. from tinguaite at Mount Royal. Although this method, applied to a whole rock containing alkali feldspar, is now known to be unreliable, the low number, as with Osborne's estimate, is a guide for more precise work.

An unpublished age of 375 m.y. on granite from Megantic was reported in 1958 from our laboratory. This granite was obtained from material near the base of the mountain which on later petrographic investigation by A. M. Reid (personal communication) proved not to be in situ. It is lithologically identical with Devonian granite from Scotstown and elsewhere in the neighborhood.

Appendix 2. List of Samples

Oka

C3253 Carbonatite, typical niobium ore, donated by J. A. Gower.
C4584 Carbonatite, collected by J. L. Powell.

Mount Royal

R3069 Tinguaite dike, St. Josef's Boulevard, Montreal East, collected by H. W. Fairbairn.
R4282 Essexite, Camelien Houde Boulevard, collected by G. H. Beall.

St. Bruno

R4704 Gabbro, collected by G. H. Beall.

St. Hilaire

R4270 Essexite, collected by G. H. Beall.
R4281 Essexite, collected by G. H. Beall.

Rougement

R4703 Peridotite, collected by G. H. Beall.

Johnson

R4180 Essexite, collected by G. H. Beall.
R4268A Pulaskite (coarse phase), collected by G. H. Beall.

Yamaska

R4179 Yamaskite, collected by G. H. Beall.
R4273A Gabbro, collected by G. H. Beall.
R4273B Syenite, collected by G. H. Beall.

Shefford

R4269A Nordmarkite, collected by G. H. Beall.
R4181 Nordmarkite, collected by G. H. Beall.

Brome

R4564 Nordmarkite, collected by G. Faure.
R4768A Essexite (dark phase), collected by J. L. Powell.
R4768B Essexite (light phase), collected by J. L. Powell.

Megantic

R4196 Granite, collected by G. H. Beall.

Appendix 3. Rock Names

Nordmarkite, a quartz-bearing alkali syenite.
Essexite, a type of alkali gabbro.
Pulaskite, porphyritic nepheline syenite.
Tinguaite, porphyritic dike equivalent of aegirine phonolite.
Yamaskite, hornblende-rich gabbro.

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